

Progress Test 1.A — Elec-S1

1.A · From potential to electric voltage
 Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 7 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Unit conversion (1 mark)

Convert the voltage $u = 0.0258\text{ MV}$ into millivolts. Give the exact result in scientific notation, without rounding.

[Empty grid for answer]

Marking

①
 0 1
 [] []

Exercise 2. Electric voltage — definition (5 marks)

Define the electric voltage u_{AB} between two points A and B of a circuit. Write down its expression in terms of the electric potentials, deduce the relation between u_{AB} and u_{BA} , and state the SI unit of voltage (name and symbol).

[Empty grid for answer]

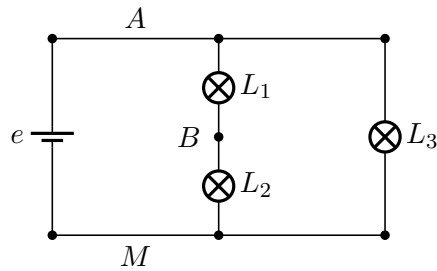
Marking

①
 0 1 2
 [] [] []
 ②
 0 1 2
 [] [] []
 ③
 0 1
 [] []

Exercise 3. Potentials in a circuit

(3 marks)

The circuit below contains an ideal voltage source of electromotive force $e = 6.0 \text{ V}$ and three lamps. The wires are ideal. The wire M , connected to the $-$ terminal of the source, is taken as the reference of potentials: $V_M = 0$.



How many distinct electric potentials are there in this circuit?

- ☐ 2 ☐ 3 ☐ 4 ☐ 7

The voltage across L_2 is $u_{BM} = 2.0 \text{ V}$. The voltage u_{AB} across L_1 is:

- ☐ 2.0 V ☐ 4.0 V ☐ 6.0 V ☐ 8.0 V

A voltmeter has its V terminal on B and its COM terminal on A . It displays:

- ☐ +4.0 V ☐ -4.0 V ☐ +2.0 V ☐ -6.0 V

Exercise 4. Potentials and voltages

(5 marks)

In a circuit, the ground M is taken as the reference of potentials ($V_M = 0$). Three points A , B and C have the potentials given below. Express, then compute, the voltages u_{AB} , u_{BC} and u_{CA} . Compute their sum and comment.

$V_A = 12.0\text{ V}$	$V_B = 4.5\text{ V}$	$V_C = -3.0\text{ V}$
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Marking

1

01

2

01

3

01

4

01

5

01

— End of the test —